



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

June 2012

Assessment Task 3
Year 12

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question if full marks are to be awarded.
- Answer in simplest exact form unless otherwise instructed.

Total Marks – 62

- Attempt sections A – C.
- Start each **NEW** section in a separate answer booklet.
- Hand in your answers in 3 separate bundles:

Section A
Section B
Section C

Examiner: *J. Chen*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

START A NEW ANSWER BOOKLET

SECTION A [20 marks]

Marks

For these 10 questions there is one correct answer per question. Write down in your answer booklet the question number and letter of your answer.

1. 1.58 radians approximately equal

[1]

- (A) $90^{\circ}31'$
- (B) $90^{\circ}32'$
- (C) $90^{\circ}53'$
- (D) None of the above

- 2.

[1]

$$\lim_{x \rightarrow 0} \frac{\tan 3x}{x}$$

equals

- (A) $\frac{1}{3}$
- (B) 3
- (C) 0
- (D) 1

- 3.

[1]

$$\int \frac{dx}{x^2 + 4x + 5}$$

equals

- (A) $\tan^{-1}\left(\frac{x+2}{2}\right) + C$
- (B) $\tan^{-1}\left(\frac{x}{2}\right) + C$
- (C) $\tan^{-1}(x + 2) + C$
- (D) $2 \tan^{-1}\left(\frac{x}{2}\right) + C$

4.

[1]

$$\int_0^{\frac{\pi}{4}} (1 + \tan^2 x) \cdot dx$$

equals

- (A) -1
- (B) 0
- (C) 1
- (D) 2

5. The domain of the function

[1]

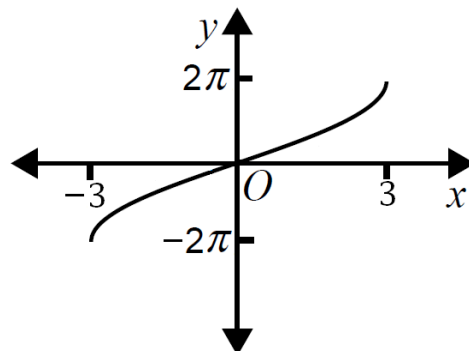
$$y = 2 \sin^{-1} \left(\frac{x}{2} \right)$$

is

- (A) $-2 \leq x \leq 2$
- (B) $-\frac{1}{2} \leq x \leq \frac{1}{2}$
- (C) $-2 \leq y \leq 2$
- (D) $-\frac{1}{2} \leq y \leq \frac{1}{2}$

6. The diagram below shows the graph of a function.

[1]



A possible equation for the function is

- (A) $y = 4 \sin^{-1}(3x)$
- (B) $y = 4 \sin^{-1} \left(\frac{x}{3} \right)$
- (C) $y = \frac{1}{4} \sin^{-1}(3x)$
- (D) $y = \frac{1}{4} \sin^{-1} \left(\frac{x}{3} \right)$

7.

[1]

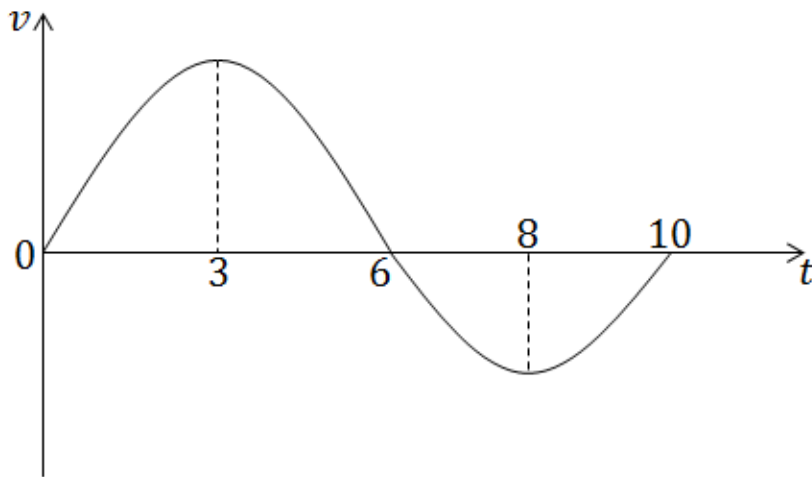
$$\int_{-2}^2 \sqrt{4-x^2} \cdot dx$$

equals

- (A) 0
- (B) 2π
- (C) 4π
- (D) $\frac{\pi}{2}$

8. The graph represents the velocity v m/s of a particle after t seconds travelling in a straight line. The particle starts from rest at the origin.

[1]



When does the particle return to the origin?

- (A) $t = 2$
- (B) $t = 4$
- (C) $t = 10$
- (D) The particle does not return to the origin.

9.

[1]

$$\frac{d}{dx} \left(\frac{\sin x}{x} \right)$$

Equals

(A) $\frac{x \cos x + \sin x}{x^2}$

(B) $\frac{x \cos x - \sin x}{x^2}$

(C) $\frac{\sin x - x \cos x}{x^2}$

(D) $\frac{\sin x + x \cos x}{x^2}$

10. How many solutions does the equation $\sin 2x = \cos x$ have in the domain $0 \leq x \leq 2\pi$?

[1]

(A) 0

(B) 1

(C) 2

(D) None of the above

End of Multiple Choice Section

11. Find the inverse function f^{-1} of the function f , defined by $f(x) = 2 \log_e x + 3$. [1]
Express the result in the form y in terms of x .

12. Find the exact value of [3]

$$\tan^{-1} \frac{1}{4} + \tan^{-1} \frac{3}{5}$$

13. State the domain and range of the function: [2]

$$f(x) = 2 \cos^{-1} \frac{x}{3}$$

14. [4]

(i) Differentiate $x \sin^{-1} x + \sqrt{1 - x^2}$.

(ii) Hence evaluate

$$\int_0^1 \sin^{-1} x \cdot dx$$

End of Section A

START A NEW ANSWER BOOKLET

SECTION B [20 marks]

Marks

1. Find $\int x\sqrt{1+x^2} \cdot dx$, using the substitution $u = 1 + x^2$.

[3]

2. Find

[3]

$$\int_0^{\pi} \sin^2 x \cdot dx$$

3. Evaluate

[2]

$$\lim_{x \rightarrow 0} \frac{\tan 6x}{\sin 4x}$$

- 4.

[4]

- (i) Show that there is a root to the equation $\sin x = x - \frac{1}{2}$ between $x = 0.5$ and $x = 1.8$.
- (ii) Taking $x = 1.2$ as a first approximation to this solution, apply Newton's method once to find a closer approximation to the solution. Give your answer correct to two decimal places.

- 5.

[4]

- (i) Write down a primitive function of $e^{f(x)} \cdot f'(x)$.
- (ii) Hence, evaluate

$$\int_{-1}^1 \frac{2e^{\cos^{-1} x}}{\sqrt{1-x^2}} \cdot dx$$

6. The surface area of a cube is increasing at a rate of 12.6 cm^2 per second and initially each edge is 10 cm.

[4]

- (i) If it maintains its cubic shape, find an expression in terms of t for the surface area of the cube after t seconds.
- (ii) Hence, or otherwise, find the rate at which the volume of the cube is increasing after 10 seconds.

End of Section B

START A NEW ANSWER BOOKLET

SECTION C [22 marks]

Marks

1. [5]
- (i) Prove that
- $$\frac{1}{x^2 + 1} - \frac{1}{x^2 + 3} = \frac{2}{(x^2 + 1)(x^2 + 3)}$$
- (ii) Hence determine the value of
- $$\int_{-\sqrt{3}}^1 \frac{dx}{(x^2 + 1)(x^2 + 3)}$$
2. [5]
- (i) Prove by Mathematical Induction that if x is a positive integer then $(x + 1)^n - 1$ is divisible by x for all positive integers $n > 0$.
- (ii) Hence, deduce that $15^n - 5^n - 3^n + 1$ is divisible by 8 for all positive integers $n > 0$.
3. [6]
- A particle moves with acceleration $\ddot{x} = 2x^3 - 10x$ where x is the displacement in metres. Initially $v = 0$ when $x = -1$.
- (i) Find v in terms of x .
- (ii) Explain why there is a restriction on x and clearly indicate the restriction.
- (iii) Describe briefly what would have happened if the initial conditions where $v = 0$ when $x = 0$.
4. [6]
- Assume that tides rise and fall in **Simple Harmonic Motion**. On June 25 the depth of water in a harbour entrance at low tide is 2 metres at 11 a.m. At the following high tide at 5:20 p.m. the depth is 6 metres. Calculate the first time period during which a yacht can safely enter the harbour if minimum depth of 3.5 metres of water is needed.

End of Section C
End of Exam

1. (B) $90^\circ 32'$

2. $\lim_{x \rightarrow 0} \frac{\tan 3x}{x}$
 $= \lim_{3x \rightarrow 0} \frac{\tan 3x}{3x} \times 3$
 $= 1 \times 3$
 $= 3$ (B)

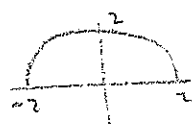
3. $\int \frac{dx}{x^2 + 4x + 5}$
 $= \int \frac{dx}{(x+2)^2 + 1}$
 $= \tan^{-1}(x+2) + C$ (C)

4. $\int_0^{\pi/4} (1 + \tan^2 x) dx$
 $= \int_0^{\pi/4} \sec^2 x dx$
 $= [\tan x]_0^{\pi/4}$
 $= \tan \frac{\pi}{4} - \tan 0$
 $= 1$ (C)

5. $y = 2 \sin^{-1} \left(\frac{x}{2} \right)$
 $-1 \leq \frac{x}{2} \leq 1$
 $-2 \leq x \leq 2$ (A)

6. $y = 4 \sin^{-1} \frac{x}{3}$ (B)

7. $\int_{-2}^2 \sqrt{4-x^2} dx$
 $= \frac{1}{2} \times \pi \times 2^2$
 $= 2\pi$ (B)



8. (D) Particle does not return to origin

9. $\frac{d}{dx} \left(\frac{\sin x}{x} \right)$
 $= \frac{x \cdot \cos x - \sin x \cdot 1}{x^2}$
 $= \frac{x \cos x - \sin x}{x^2}$ (B)

10. $\sin 2x = \cos x$

$\therefore 2 \sin x \cos x = \cos x$

$\therefore \cos x (2 \sin x - 1) = 0$

$\therefore \cos x = 0$ or $\sin x = \frac{1}{2}$

$\therefore x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$

4 solutions (D)
 (None of the above)

11. $x = 2 \log_e y + 3$

$x - 3 = 2 \log_e y$

$\log y = \frac{x-3}{2}$

$y = e^{\frac{x-3}{2}}$

(1)

$$12. \quad x = \tan^{-1} \frac{1}{4} + \tan^{-1} \frac{3}{5}$$

$$\tan x = \tan \left(\tan^{-1} \frac{1}{4} + \tan^{-1} \frac{3}{5} \right)$$

$$= \frac{\tan \left(\tan^{-1} \frac{1}{4} \right) + \tan \left(\tan^{-1} \frac{3}{5} \right)}{1 - \tan \left(\tan^{-1} \frac{1}{4} \right) \tan \left(\tan^{-1} \frac{3}{5} \right)}$$

$$= \frac{\frac{1}{4} + \frac{3}{5}}{1 - \frac{1}{4} \times \frac{3}{5}}$$

$$= \frac{\left(\frac{17}{20} \right)}{\left(\frac{17}{20} \right)}$$

$$= 1$$

③

$$\therefore x = \frac{\pi}{4}$$

$$13. \quad f(x) = 2 \cos^{-1} \frac{x}{3}$$

$$\text{Domain} \quad -1 \leq \frac{x}{3} \leq 1$$

$$\therefore -3 \leq x \leq 3$$

①

$$\text{Range} \quad 0 \leq \cos^{-1} \frac{x}{3} \leq \pi$$

$$0 \leq 2 \cos^{-1} \frac{x}{3} \leq 2\pi$$

$$\therefore 0 \leq y \leq 2\pi$$

①

$$14. (i) \quad \frac{d}{dx} \left(x \sin^{-1} x + \sqrt{1-x^2} \right)$$

$$= \sin^{-1} x \cdot 1 + x \cdot \frac{1}{\sqrt{1-x^2}} + \frac{1}{2} \cdot \frac{1}{\sqrt{1-x^2}} \cdot x \cdot (-2x)$$

②

$$= \sin^{-1} x$$

$$(ii) \quad \int_0^1 \sin^{-1} x \, dx = \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^1$$

$$= [1 \cdot \sin^{-1} 1 + 0] - [0 + 1]$$

$$= \frac{\pi}{2} - 1$$

②

2012 Extension 1 Mathematics Task 3:
Solutions— Section B

1. Find $\int x\sqrt{1+x^2}.dx$, using the substitution $u = 1 + x^2$.

3

Solution: $\frac{du}{dx} = 2x,$
 $I = \frac{1}{2} \int \frac{du}{dx} \cdot u^{\frac{1}{2}} dx,$
 $= \frac{1}{2} \cdot u^{\frac{3}{2}} \cdot \frac{2}{3} + c,$
 $= \frac{(1+x^2)\sqrt{1+x^2}}{3} + c.$

2. Find

$$\int_0^{\pi} \sin^2 x . dx$$

3

Solution: $I = \frac{1}{2} \int_0^{\pi} (1 - \cos 2x) dx,$
 $= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi},$
 $= \frac{1}{2} \{ \pi - 0 - (0 - 0) \},$
 $= \frac{\pi}{2}.$

$\cos 2x = 1 - 2 \sin^2 x,$
 $\sin^2 x = \frac{1 - \cos 2x}{2}.$

3. Evaluate

$$\lim_{x \rightarrow 0} \frac{\tan 6x}{\sin 4x}$$

2

Solution: First method—

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan 6x}{\sin 4x} &= \lim_{x \rightarrow 0} \frac{\tan 6x}{6x} \cdot \frac{4x}{\sin 4x} \cdot \frac{6}{4}, \\ &= 1 \times 1 \times \frac{6}{4}, \\ &= \frac{3}{2}. \end{aligned}$$

Solution: Second method—

$$\begin{aligned}
 \frac{\tan 6x}{\sin 4x} &= \frac{\sin 6x}{\cos 6x \cdot \sin 4x}, \\
 &= \frac{\sin 4x \cdot \cos 2x + \cos 4x \cdot \sin 2x}{\cos 6x \cdot \sin 4x}, \\
 &= \frac{\sin 4x \cdot \cos 2x}{\cos 6x \cdot \sin 4x} + \frac{\cos 4x \cdot \sin 2x}{\cos 6x \times 2 \sin 2x \cdot \cos 2x}, \\
 &= \frac{\cos 2x}{\cos 6x} + \frac{\cos 4x}{2 \cos 6x \cdot \cos 2x} \\
 \therefore \lim_{x \rightarrow 0} \frac{\tan 6x}{\sin 4x} &= \frac{1}{1} + \frac{1}{2 \times 1 \times 1}, \\
 &= \frac{3}{2}.
 \end{aligned}$$

4. (a) Show that there is a root to the equation $\sin x = x - \frac{1}{2}$ between $x = 0.5$ and $x = 1.8$.

4

Solution: Put $f(x) = x - \sin x - \frac{1}{2}$,
 $f(0.5) \approx -0.48$,
 $f(1.8) \approx 0.33$.

$\therefore$ There is a root where $0.5 < x < 1.8$.

- (b) Taking $x = 1.2$ as a first approximation to this solution, apply Newton's method once to find a closer approximation to the solution. Give your answer correct to two decimal places.

Solution: $f'(x) = 1 - \cos x$,
 $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$,
 $= 1.2 - \frac{1.2 - \sin 1.2 - 0.5}{1 - \cos 1.2}$,
 ≈ 1.56 (2 dec. pl.)

5. (a) Write down a primitive function of $e^{f(x)} \cdot f'(x)$.

4

Solution: $e^{f(x)}$ or $e^{f(x)} + c$.

(b) Hence evaluate

$$\int_{-1}^1 \frac{2e^{\cos^{-1} x}}{\sqrt{1-x^2}} dx$$

Solution:
$$I = -2 \int_{-1}^1 \frac{-1 \times e^{\cos^{-1} x}}{\sqrt{1-x^2}} dx,$$

$$= -2 \left[e^{\cos^{-1} x} \right]_{-1}^1,$$

$$= -2(1 - e^{\pi}),$$

$$= 2(e^{\pi} - 1).$$

6. The surface area of a cube is increasing at a rate of 12.6 cm^2 per second and initially each edge is 10 cm .

4

- (a) If it maintains its cubic shape, find an expression in terms of t for the surface area of the cube after t seconds.

Solution: Let edge length be x , surface area be a , and volume be v .

$$\frac{da}{dt} = 12.6,$$

$$a = \int 12.6 dt,$$

$$= 12.6t + c.$$

When $t = 0$, $a = 6 \times 10^2 = 600$,

$$\therefore a = 12.6t + 600.$$

- (b) Hence or otherwise, find the rate at which the volume of the cube is increasing after 10 seconds.

Solution: First method—

$$v = x^3, \quad a = 6x^2,$$

$$\frac{dv}{dx} = 3x^2, \quad \frac{da}{dx} = 12x.$$

$$\frac{dv}{dt} = \frac{dv}{dx} \times \frac{dx}{da} \times \frac{da}{dt},$$

$$= 3x^2 \times \frac{1}{12x} \times 12.6,$$

$$= 3.15x.$$

When $t = 10$,

$$6x^2 = 12.6t + 600,$$

$$= 726,$$

$$x^2 = 121,$$

$$x = 11 \text{ as } x > 0.$$

$$\therefore \frac{dv}{dt} = 3.15 \times 11,$$

$$= 34.65.$$

So the volume is increasing at $34.65 \text{ cm}^3/\text{s}$.

Solution: Second method—

$$6x^2 = 12.6t + 600,$$

$$x^2 = 2.1t + 100,$$

$$x^3 = (2.1t + 100)^{3/2},$$

$$v = (2.1t + 100)^{3/2},$$

$$\frac{dv}{dt} = 2.1 \times \frac{3}{2}(2.1t + 100)^{1/2},$$

$$= 3.15\sqrt{2.1t + 100}.$$

When $t = 10$,

$$\frac{dv}{dt} = 3.15\sqrt{2.1(10) + 100},$$

$$= 34.65.$$

So the volume is increasing at $34.65 \text{ cm}^3/\text{s}$.

Solution: Third method—

$$\frac{dv}{dt} = \frac{dv}{da} \times \frac{da}{dt}.$$

$$a = 6x^2,$$

$$v = x^3,$$

$$x = \sqrt{\frac{a}{6}}.$$

$$= \sqrt{\frac{a^3}{6^3}},$$

$$= \frac{a^{3/2}}{6^{3/2}}.$$

$$= \frac{6\sqrt{6}}{3} \cdot \frac{a^{1/2}}{a^{1/2}}.$$

$$\frac{dv}{da} = \frac{2}{2} \cdot \frac{6\sqrt{6}}{2\sqrt{6}},$$

$$= \frac{\sqrt{a}}{4\sqrt{6}}.$$

$$\text{When } t = 10, a = 726, \therefore \frac{dv}{dt} = \frac{\sqrt{726}}{4\sqrt{6}} \times 12.6,$$

$$= \frac{11}{4} \times 12.6,$$

$$= 34.65.$$

So the volume is increasing at $34.65 \text{ cm}^3/\text{s}$.

Section C

1. (i) LHS = $\frac{(x^2+3)-(x^2+1)}{(x^2+1)(x^2+3)}$

$$= \frac{2}{(x^2+1)(x^2+3)} \quad [2]$$

= RHS QED

(ii) $\int_{-\sqrt{3}}^1 \frac{dx}{(x^2+1)(x^2+3)}$

$$= \frac{1}{2} \int_{-\sqrt{3}}^1 \left(\frac{1}{x^2+1} - \frac{1}{x^2+3} \right) dx$$

$$= \frac{1}{2} \left[\tan^{-1} x - \frac{1}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} \right]_{-\sqrt{3}}^1$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{4} - \frac{1}{\sqrt{3}} \cdot \frac{\pi}{6} \right) - \left(-\frac{\pi}{3} - \frac{1}{\sqrt{3}} \left(-\frac{\pi}{4} \right) \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{4} - \frac{\pi}{6\sqrt{3}} + \frac{\pi}{3} - \frac{\pi}{4\sqrt{3}} \right]$$

$$= \frac{\pi}{2} \left(\frac{7}{12} - \frac{4+6}{24\sqrt{3}} \right)$$

$$= \frac{\pi}{2} \left(\frac{7}{12} - \frac{5}{12\sqrt{3}} \right)$$

$$= \frac{\pi}{2} \left(\frac{7\sqrt{3}-5}{12\sqrt{3}} \right) \quad [3]$$

2. (i) $p(n): x | (n+1)^n - 1$ if $x \in \mathbb{Z}^+$

$p(1)$: If $n=1$, $(n+1)^n - 1$ is divisible by x which is divisible by x .

$\therefore p(1)$ is true.

$p(k)$: Assume true for $n=k$

i.e. $x | (k+1)^k - 1$
 $\therefore (k+1)^k - 1 = Rx$, some $R \in \mathbb{Z}$

$p(k+1)$: RTP that $p(k) \rightarrow p(k+1)$

Now when $n=k+1$

$$(x+1)^{k+1} - 1$$

$$= (x+1)^k (x+1) - 1$$

$$= (Rx+1)(x+1) - 1$$

by assumption

$$= Rx^2 + Rx + x + 1 - 1$$

$$= x(Rx + R + 1)$$

which is divisible by x .

$\therefore$ Since $p(1)$ is true

and $p(k) \rightarrow p(k+1)$

then by PMI, $p(n)$ is true. [3]

(ii) $15^n - 5^n - 3^n + 1$

$$= 5^n(3^n - 1) - (3^n - 1)$$

$$= (5^n - 1)(3^n - 1)$$

$$= 4R \cdot 2S \text{ (say) } RSE$$

$$= 8(RS) \quad \text{QED [2]}$$

3. $\ddot{x} = 2x^3 - 10x$

$v=0, x=-1$ when $t=0$

(i) $\frac{d}{dx}(\frac{1}{2}v^2) = 2x^3 - 10x$

$\therefore \frac{1}{2}v^2 = \int (2x^3 - 10x) dx + C$
 $= \frac{2x^4}{4} - \frac{10x^2}{2} + C$

$v^2 = x^4 - 10x^2 + C'$
 $v=0$ when $x=-1$, so

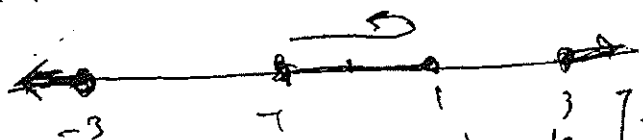
$0 = 1 - 10 + C'$

$C' = 9$

So $v^2 = x^4 - 10x^2 + 9$

$v = \pm \sqrt{x^4 - 10x^2 + 9}$
 $= \pm \sqrt{(x^2-9)(x^2-1)}$ [3]

(ii) $x^2 \leq 1$ or $x^2 \geq 9$
 $(x^2-9)(x^2-1) \geq 0$
 $\therefore -1 \leq x \leq 1$ or $x \leq -3$ or $x \geq 3$



ie Since particle starts at -1, it is limited to $-1 \leq x \leq 1$ [2]

(iii) If $v=0$ when $x=0$ then the acceleration is zero, so the particle does not move.

[1]

4.

1100 2m low

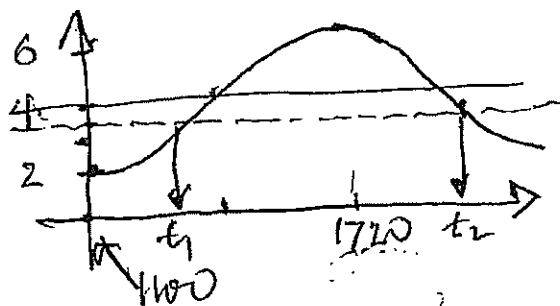
1720 6m High

+620

2340

low

$a=2$



Period $12:40 = \frac{760}{n}$ min.
 $= 12\frac{2}{3}$ hrs

"Equilibrium" $= x=4$ m

$(760 = \frac{2\pi}{n})$ $12\frac{2}{3} = \frac{2\pi}{n}$

$n = \frac{2\pi}{760}$ $n = \frac{2\pi}{12\frac{2}{3}}$

$(= 0.008267) = 2.49604$

Hence $x = -2 \cos(nt) + 4$

For $x = 3.5$ m

$3.5 = -2 \cos(nt) + 4$

$-0.5 = -2 \cos(nt)$

$+0.25 = \cos(6t)$

$nt = \cos^{-1}(0.25)$

$t_1 = 2h 39' + 11:00$

$= 13:39$

$t_2 = 23:40 - 2:39$

$= 21:01$

[6]